

Charlie Marx

CURRICULUM VITAE - JULY 2020

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EDUCATION	Stanford University Ph.D. Student in Computer Science	2020 - PRESENT
	Haverford College B.S. in Computer Science, <i>high honors</i> B.S. in Mathematics, <i>high honors</i> GPA: 3.96/4.0, <i>magna cum laude</i>	2016 - 2020
AWARDS & HONORS	Phi Beta Kappa CRA Outstanding Undergraduate Researcher Runner-Up Beckman Scholars Award \$26,000 Farmers Insurance Standout Student of the Year \$2,500	2020 2019 2018 2016
PUBLICATIONS	MACHINE LEARNING METHODS 1. Predictive Multiplicity in Classification Charles T. Marx , Flavio du Pin Calmon, Berk Ustun ICML – <i>International Conference on Machine Learning</i> , 2020 Previously Selected for Oral Presentation at 2019 NeurIPS HCML Workshop 2. Disentangling Influence: Using Disentangled Representations to Audit Model Predictions Charles T. Marx , Richard L. Phillips, Sorelle A. Friedler, Carlos Scheidegger, Suresh Venkatasubramanian NeurIPS – <i>Conference on Neural Information Processing Systems</i> , 2019 MACHINE LEARNING APPLICATIONS 3. Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill Helen White, Charles T. Marx , David Valentine, Charles Sharpless, Christoph Aeppli, Kelsey Gosselin, Veronika Kivenson, Rachel Liu, Robert Nelson, Sean Sylva, Christopher Reddy <i>ACS Earth and Space Chemistry</i> , 2019 4. Rapid Identification of Deepwater Horizon Oil Residues Using X-Ray Fluorescence Anna Michel, Alexandra E. Morrison, Charles T. Marx , Helen White <i>ACS Environmental Science & Technology Letters</i> , 2018	
RESEARCH EXPERIENCE	Research Assistant in Machine Learning <i>Harvard University</i> Studied predictive multiplicity: the ability for datasets to admit multiple equally accurate models with conflicting predictions. Developed integer programming tools and software to measure the severity of multiplicity. Compiled findings into first-author conference and workshop paper. <i>Advisors:</i> Flavio Calmon and Berk Ustun Research Assistant in Machine Learning <i>Haverford College</i> Designed new methods to quantify indirect feature influence in machine learning models using disentangled representations and adversarial learning. <i>Advisor:</i> Sorelle Friedler Research Assistant in Chemistry <i>Haverford College</i> Designed machine learning tools to optimize model performance, robustness, and interpretability for on-site oil identification. <i>Advisor:</i> Helen White Summer Research Intern <i>Prince William Sound Science Center</i> Automated salmon and herring tracking in Alaska using machine learning and sonar data. <i>Advisor:</i> Scott Pegau	SUMMER 2019 2017 – 2019 2017 – 2018 SUMMER 2018

ACADEMIC ACTIVITIES	Teaching Assistant in Machine Learning	2019 – 2020
	Held weekly office hours and graded for an upper-level machine learning course.	
	Peer Tutor	2018 – 2019
	Tutored students in statistics, linear algebra, and machine learning.	
	Computer Science Faculty Search Committee	2018 – 2019
	Student representative on the computer science tenure-track faculty search committee.	
INVITED & CONTRIBUTED TALKS	Simons Institute for the Theory of Computing <i>Berkeley, CA</i>	2019
	Disentangling Influence (45-minute talk)	
	INFORMS Annual Meeting <i>Seattle, WA</i>	2019
	Predictive Multiplicity in Classification (20-minute talk)	
	FAT* Conference <i>New York, NY</i>	2018
	Auditing Black Box Models (60-minute tutorial) <i>with Carlos Scheidegger and Suresh Venkatasubramanian</i>	
	GOMOSSES Conference <i>New Orleans, LA</i>	2018
	Examining Inputs of Biogenic and Oil-Derived Hydrocarbons (15-minute talk)	
POSTER PRESENTATIONS	ICML Conference <i>Vienna, Austria</i>	2020
	Predictive Multiplicity in Classification	
	NeurIPS Conference <i>Vancouver, Canada</i>	2019
	Disentangling Influence: Using Disentangled Representations to Audit Model Predictions	
	NeurIPS Workshop on Human-Centric Machine Learning <i>Vancouver, Canada</i>	2019
	On the Multiplicity of Predictions in Machine Learning	
	Beckman National Symposium <i>Irvine, CA</i>	2019
	Indirect Influence in Machine Learning Models	
	Beckman Regional Symposium <i>New York, NY</i>	2018
	Mitigating Social Discrimination in Deep Learning	
	KINSC Undergraduate Research Symposium <i>Haverford, PA</i>	2017
	Examining Biogenic Hydrocarbons After the Deepwater Horizon Oil Spill	
FIELDWORK	Alaska Prince William Sound Research Charter	2018
	Gulf of Mexico Shoreline Research Tour	2017
CONFERENCE REVIEWING	KDD – International Conference on Knowledge Discovery and Data Mining	2020
EXTRA– CURRICULAR ACTIVITIES	Captain of Haverford College Men’s Club Soccer	2018 – 2019
	Leader of Haverford College Outdoors Club	2017 – 2018
	President of Furniture for Newly Housed Families Organization	2015 – 2017
SKILLS & INTERESTS	Languages: <i>Fluent:</i> Python, R, LaTeX · <i>Familiar:</i> SQL, C, Unix	
	Tools: Keras/Tensorflow, PyTorch, Pandas, CPLEX	
	Interests: Backpacking, Gardening, Soccer, Folk Music, Cooking	